

Bachelor of Science in Mathematics — Program Handbook

Self-Taught University · Version 1.0 · October 2026

Program Overview

The Bachelor of Science in Mathematics is a self-directed undergraduate program of 32 courses and 5,440 nominal hours, modeled on the structure of MIT Course 18 (Mathematics). It is designed for adults who want a complete mathematics education studied at their own pace, with the record of their work open for anyone to examine.

Item	Value
Format	Self-directed study with published textbooks and open reference courses
Courses required	32: 17 fixed, 11 electives, and 4 free electives, with 2 of the electives taken as seminars
Nominal hours	170 per course (equivalent to a 12-unit MIT subject); 5,440 in total
Reference pace	2 courses per 4-month term, about 20 hours of study per week
Reference length	16 terms, about 5 years and 4 months
Language	English
Cost	None for the program; textbooks as chosen by the student

The program is inspired by MIT Course 18 and uses MIT OpenCourseWare and other university materials as reference courses. It is not affiliated with, endorsed by, or connected to MIT or any other institution. Completing it does not confer an accredited degree; the credential rests on the work the student publishes (see Credential and Verification).

Mission and Program Learning Outcomes

The program's mission is to give any motivated adult the mathematical education of a strong undergraduate major, and to replace institutional endorsement with openly verifiable work.

Graduates of the program are able to:

1. Read, write, and critique rigorous proofs in analysis, algebra, topology, and discrete mathematics.
2. Use the core tools of single and multivariable calculus, linear algebra, and differential equations to model and solve problems.
3. Reason about uncertainty with probability and statistics, from definitions to inference.
4. Work with the central structures of modern mathematics: groups, rings, fields, vector spaces, metric and topological spaces.
5. Select and justify methods in at least one area of depth chosen from the electives.
6. Apply mathematics to physical systems and to computation.

7. Communicate mathematics in writing and in an oral presentation to a mathematically literate audience.
8. Study independently from primary texts, plan long-term work, and document it so that others can verify it.

Degree Requirements

The program follows the blocks of an MIT bachelor's degree. At MIT, every student takes 17 institute requirements, the subjects of the major, and free electives. Here the 17 fixed courses keep the subjects every MIT mathematics major takes and replace the institute requirements outside the field (chemistry, biology, humanities) with mathematics and computing, so the whole degree stays in the field.

Requirement	Courses	Notes
Fixed courses	17	The same for every student. Nine are taken in some form by every MIT mathematics major: single and multivariable calculus, physics I and II, differential equations, linear algebra, real analysis, probability and statistics, and abstract algebra I. Two more are in mathematics: complex variables and topology. Six are in computing with mathematical content: mathematics for computer science, programming in Python, introduction to algorithms, theory of computation, cryptography, and mathematical logic
Electives	11	Chosen by the student from the mathematics electives and topics courses
Free electives	4	Chosen from the whole catalog; any course can be counted as a free elective
Seminars	2	Topics courses taken as a seminar, assessed by a recorded presentation and a paper; they still count as electives
Total	32	5,440 nominal hours

- **Classification.** Courses follow MIT's classification of their reference subjects: subjects of MIT's mathematics department are electives, subjects of other departments are free electives, and communication-intensive subjects can be taken as seminars. Any course can be counted as a free elective, so no course is wasted.
- **Course weight.** Every course carries 170 nominal hours, the workload of a 12-unit MIT subject (about 12 hours per week for 14 weeks).
- **Prerequisites.** Each course lists its prerequisites. Taking a course before its prerequisites is allowed but flagged; the student is responsible for any gap.
- **Course numbers.** 100-level courses are introductory, 200-level intermediate, 300-level advanced, and 400-level topics courses and seminars. MATH, CS, and PHYS mark the field.

Program Structure

The reference plan takes 2 courses per 4-month term, 3 terms per year. In the first year (terms 1–3) both courses are fixed. From the second year on, as at MIT, each term has one fixed course and one course the student chooses, so electives and seminars start early and run alongside the fixed courses. Terms are flexible: each term starts when the previous one ends, and finishing early or late moves every later term with it.

Term	Fixed course	Second course
1	MATH 101 Single Variable Calculus	MATH 110 Mathematics for Computer Science

Term	Fixed course	Second course
2	MATH 102 Multivariable Calculus	MATH 120 Linear Algebra
3	CS 101 Introduction to Programming in Python	PHYS 101 Classical Mechanics
4	MATH 130 Differential Equations	Chosen by the student
5	MATH 210 Real Analysis	Chosen by the student
6	MATH 221 Abstract Algebra I	Chosen by the student
7	PHYS 102 Electricity and Magnetism	Chosen by the student
8	CS 120 Introduction to Algorithms	Chosen by the student
9	MATH 140 Probability and Statistics	Chosen by the student
10	MATH 370 Mathematical Logic and Set Theory	Chosen by the student
11	CS 340 Theory of Computation	Chosen by the student
12	MATH 230 Complex Variables	Chosen by the student
13	CS 350 Cryptography	Chosen by the student
14	MATH 260 Topology	Chosen by the student
15–16	Chosen by the student	Chosen by the student

At 170 hours per course, two courses in a 4-month term require about 20 hours of study per week. Students who study fewer hours can lengthen the terms or take one course at a time; the requirements do not change.

Course Catalog: Fixed Courses

Every course carries 170 hours. Texts list the primary text first, with chapters where they apply; check chapters against your edition.

MATH 101 Single Variable Calculus

Prerequisites: none. Limits and continuity; derivatives and their applications to rates, approximation, and optimization; the definite integral and the fundamental theorem of calculus; techniques and applications of integration; sequences and series, including power and Taylor series.

Texts: Stewart, Clegg, and Watson, *Calculus: Early Transcendentals*, 9th ed., ch. 1–11; Hass, Heil, and Weir, *Thomas' Calculus*, 14th ed. Reference: [MIT 18.01](#).

MATH 110 Mathematics for Computer Science

Prerequisites: none. Proofs by induction, contradiction, and construction; sets, relations, functions, and logic; number theory and modular arithmetic; graphs and trees; counting, generating functions, and recurrences; discrete probability.

Texts: Lehman, Leighton, and Meyer, *Mathematics for Computer Science*, Parts I–III; Velleman, *How to Prove It*, 3rd ed., ch. 1–6. Reference: [MIT 18.062J](#) / [6.042J](#).

CS 101 Introduction to Programming in Python

Prerequisites: none. The introduction to computer science and algorithms through programming: variables, control flow, and functions; strings, lists, and dictionaries; recursion; object-oriented programming; testing and debugging; searching and sorting, and an introduction to algorithmic complexity.

Texts: Guttag, *Introduction to Computation and Programming Using Python*, 3rd ed.; Downey, *Think Python*, 2nd ed. Reference: [MIT 6.0001](#).

PHYS 101 Classical Mechanics

Prerequisites: MATH 101. Kinematics; Newton's laws; work and energy; momentum and collisions; rotational motion and angular momentum; gravitation; oscillations.

Texts: Halliday, Resnick, and Walker, *Fundamentals of Physics*, 10th ed., ch. 1–13, 15; Kleppner and Kolenkow, *An Introduction to Mechanics*, 2nd ed., ch. 1–7. Reference: [MIT 8.01SC](#).

MATH 102 Multivariable Calculus

Prerequisites: MATH 101. Vectors and the geometry of space; partial derivatives, gradients, and optimization with constraints; double and triple integrals; vector fields, line and surface integrals; the theorems of Green, Stokes, and Gauss.

Texts: Stewart et al., ch. 12–16; Hass et al. Reference: [MIT 18.02](#).

MATH 120 Linear Algebra

Prerequisites: MATH 101. Systems of linear equations and elimination; matrix algebra; the four fundamental subspaces; orthogonality, projections, and least squares; determinants; eigenvalues and diagonalization; symmetric and positive definite matrices; the singular value decomposition.

Texts: Strang, *Introduction to Linear Algebra*, 6th ed., ch. 1–7; Lang, *Introduction to Linear Algebra*, 2nd ed. Reference: [MIT 18.06](#).

PHYS 102 Electricity and Magnetism

Prerequisites: PHYS 101, MATH 102. Electric fields and Gauss's law; electric potential; capacitance and circuits; magnetic fields; induction; Maxwell's equations and electromagnetic waves.

Texts: Halliday, Resnick, and Walker, ch. 21–33; Purcell and Morin, *Electricity and Magnetism*, 3rd ed., ch. 1–9. Reference: [MIT 8.02](#).

MATH 130 Differential Equations

Prerequisites: MATH 101. First-order equations and their models; linear equations of higher order; the Laplace transform; systems of linear equations and phase portraits; nonlinear systems and stability; series solutions and an introduction to Fourier series.

Texts: Boyce, DiPrima, and Meade, *Elementary Differential Equations and Boundary Value Problems*, 11th ed., ch. 1–7, 9; Simmons, *Differential Equations with Applications and Historical Notes*, 3rd ed. Reference: [MIT 18.03](#).

CS 120 Introduction to Algorithms

Prerequisites: MATH 110, CS 101. Asymptotic analysis; sorting; data structures, including heaps, hash tables, and balanced search trees; graph search; shortest paths; dynamic programming.

Texts: Cormen, Leiserson, Rivest, and Stein, *Introduction to Algorithms*, 4th ed., ch. 1–4, 6–8, 10–14, 20, 22; Erickson, *Algorithms*. Reference: [MIT 6.006](#).

MATH 140 Probability and Statistics

Prerequisites: MATH 102. Probability spaces, conditional probability, and Bayes' theorem; discrete and continuous random variables; expectation, variance, and the law of large numbers; the central limit theorem; estimation, confidence intervals, and hypothesis testing; Bayesian updating.

Texts: DeGroot and Schervish, *Probability and Statistics*, 4th ed., ch. 1–9; Ross, *A First Course in Probability*, 10th ed. Reference: [MIT 18.05](#).

MATH 210 Real Analysis

Prerequisites: MATH 102, MATH 110. The real numbers and completeness; sequences and series; limits, continuity, and uniform continuity; differentiation; the Riemann integral; sequences and series of functions and uniform convergence; metric spaces.

Texts: Abbott, *Understanding Analysis*, 2nd ed., ch. 1–7; Rudin, *Principles of Mathematical Analysis*, 3rd ed., ch. 1–7. Reference: [MIT 18.100](#).

MATH 370 Mathematical Logic and Set Theory

Prerequisites: MATH 110. Propositional and first-order logic; soundness and completeness; compactness and models; undecidability and the incompleteness theorems; axiomatic set theory, ordinals, and cardinals.

Texts: Enderton, *A Mathematical Introduction to Logic*, 2nd ed., ch. 1–3; Enderton, *Elements of Set Theory*. Reference: [MIT 18.510](#) (not on OpenCourseWare).

MATH 221 Abstract Algebra I

Prerequisites: MATH 120, MATH 110. Groups, subgroups, and homomorphisms; cyclic, permutation, and quotient groups; group actions and the Sylow theorems; rings, ideals, and fields; polynomial rings.

Texts: Fraleigh, *A First Course in Abstract Algebra*, 7th ed., Parts I–IV; Artin, *Algebra*, 2nd ed., ch. 1–7. Reference: [MIT 18.701](#).

CS 340 Theory of Computation

Prerequisites: MATH 110. Finite automata and regular languages; context-free grammars; Turing machines; decidability and reducibility; time complexity, P, NP, and NP-completeness.

Texts: Sipser, *Introduction to the Theory of Computation*, 3rd ed., ch. 0–5, 7; Hopcroft, Motwani, and Ullman, *Introduction to Automata Theory, Languages, and Computation*. Reference: [MIT 18.404J](#).

MATH 230 Complex Variables

Prerequisites: MATH 102, MATH 130. Complex numbers and analytic functions; the Cauchy–Riemann equations; contour integration and Cauchy's theorem; Taylor and Laurent series; residues and their use in evaluating integrals; conformal mappings.

Texts: Saff and Snider, *Fundamentals of Complex Analysis*, 3rd ed., ch. 1–6; Stein and Shakarchi, *Complex Analysis*, ch. 1–3. Reference: [MIT 18.04](#).

CS 350 Cryptography

Prerequisites: MATH 221. Classical ciphers; discrete logarithms and Diffie–Hellman; RSA and factorization; elliptic curve cryptography; digital signatures; the security notions behind them.

Texts: Hoffstein, Pipher, and Silverman, *An Introduction to Mathematical Cryptography*, 2nd ed., ch. 1–4, 6, 7; Paar and Pelzl, *Understanding Cryptography*. Reference: [Brown Math 158](#).

MATH 260 Topology

Prerequisites: MATH 210. Topological spaces, bases, and continuous maps; product, subspace, and quotient topologies; connectedness and compactness; countability and separation axioms; an introduction to the fundamental group.

Texts: Munkres, *Topology*, 2nd ed., ch. 1–4, 9; Mendelson, *Introduction to Topology*, 3rd ed. Reference: [MIT 18.901](#).

Course Catalog: Electives

The 21 electives below are grouped by area; the ten topics courses in the next section are electives as well. Any of them can be counted as a free elective instead. Each carries 170 hours.

Algebra and Number Theory

MATH 220 Theoretical Linear Algebra. Prerequisites: MATH 120, MATH 110. Abstract vector spaces and linear maps; dimension and duality; eigenvalues and invariant subspaces; inner product spaces; the spectral theorem; operators on complex and real spaces; determinants and canonical forms. Texts: Axler, *Linear Algebra Done Right*, 4th ed., ch. 1–9; Strang, *Introduction to Linear Algebra*. Reference: [MIT 18.700](#).

MATH 222 Abstract Algebra II. Prerequisites: MATH 221, MATH 220. Rings and modules; factorization in integral domains; modules over principal ideal domains; field extensions; Galois theory and its applications. Texts: Dummit and Foote, *Abstract Algebra*, 3rd ed., ch. 4, 7–10, 13, 14; Artin, *Algebra*. Reference: [MIT 18.702](#).

MATH 250 Number Theory. Prerequisites: MATH 110. Divisibility and primes; congruences and the Chinese remainder theorem; primitive roots; quadratic residues and quadratic reciprocity; arithmetic functions; selected Diophantine equations. Texts: Niven, Zuckerman, and Montgomery, *An Introduction to the Theory of Numbers*, 5th ed., ch. 1–5; Burton, *Elementary Number Theory*, 7th ed., ch. 1–9. Reference: [MIT 18.781](#).

Analysis

MATH 310 Measure and Integration. Prerequisites: MATH 210. Outer measure and Lebesgue measure; measurable functions; the Lebesgue integral and the convergence theorems; product measures and Fubini's theorem; L^p spaces. Texts: Axler, *Measure, Integration & Real Analysis*, ch. 1–5, 7; Bartle, *The Elements of Integration and Lebesgue Measure*. Reference: [MIT 18.102](#).

MATH 311 Functional Analysis. Prerequisites: MATH 310, MATH 220. Normed, Banach, and Hilbert spaces; bounded linear operators; the Hahn–Banach, open mapping, and uniform boundedness theorems; compact and self-adjoint operators and their spectral theory. Texts: Kreyszig, *Introductory Functional Analysis with Applications*, ch. 2–4, 7, 9; Lax, *Functional Analysis*. Reference: [MIT 18.102](#).

MATH 312 Fourier Analysis. Prerequisites: MATH 210. Fourier series and their convergence; summability methods; the Fourier transform on the real line and in several variables; applications to differential equations and number theory; the finite Fourier transform. Texts: Stein and Shakarchi, *Fourier Analysis: An Introduction*, ch. 1–6; Körner, *Fourier Analysis*. Reference: [MIT 18.103](#).

MATH 313 Analysis on Manifolds. Prerequisites: MATH 210, MATH 220. Differentiation in several variables; the inverse and implicit function theorems; integration in $\mathbb{R}^n$ and change of variables; manifolds in $\mathbb{R}^n$; differential forms and the generalized Stokes' theorem. Texts: Munkres, *Analysis on Manifolds*, ch. 1–8; Spivak, *Calculus on Manifolds*. Reference: [MIT 18.101](#) (on OpenCourseWare as Analysis II).

Geometry and Topology

MATH 321 Algebraic Geometry. Prerequisites: MATH 222. Affine varieties and ideals; Gröbner bases and the division algorithm; elimination theory; the Nullstellensatz; projective varieties. Texts: Cox, Little, and O'Shea, *Ideals, Varieties, and Algorithms*, 4th ed., ch. 1–5, 8; Reid, *Undergraduate Algebraic Geometry*. Reference: [Notre Dame Math 40510](#).

MATH 360 Differential Geometry. Prerequisites: MATH 210, MATH 120. Curves in the plane and in space; curvature and torsion; regular surfaces; the first and second fundamental forms; Gaussian curvature and the Theorema Egregium; geodesics. Texts: do Carmo, *Differential Geometry of Curves and Surfaces*, 2nd ed., ch. 1–4; Pressley, *Elementary Differential Geometry*. Reference: [MIT 18.950](#).

MATH 361 Algebraic Topology. Prerequisites: MATH 260, MATH 221. Homotopy and the fundamental group; covering spaces; the Seifert–van Kampen theorem; simplicial and singular homology; applications to classical results such as the Brouwer fixed point theorem. Texts: Kosniowski, *A First Course in Algebraic Topology*; Hatcher, *Algebraic Topology*, ch. 0–2. Reference: [MIT 18.904 / 18.905](#).

Applied Mathematics

MATH 330 Partial Differential Equations. Prerequisites: MATH 130, MATH 102. The wave, heat, and Laplace equations; separation of variables and Fourier series; boundary value problems; maximum principles; Green's functions. Texts: Strauss, *Partial Differential Equations: An Introduction*, 2nd ed., ch. 1–6; Farlow, *Partial Differential Equations for Scientists and Engineers*. Reference: [MIT 18.303](#).

MATH 331 Dynamical Systems. Prerequisites: MATH 130, MATH 120. One- and two-dimensional flows; fixed points, stability, and bifurcations; limit cycles; chaos in the Lorenz system; one-dimensional maps and period

doubling. Texts: Strogatz, *Nonlinear Dynamics and Chaos*, 2nd ed., ch. 2–10; Hirsch, Smale, and Devaney, *Differential Equations, Dynamical Systems, and an Introduction to Chaos*. Reference: [MIT 18.353J](#).

MATH 332 Numerical Analysis. Prerequisites: MATH 130, MATH 120. Floating-point error; roots of equations; solution of linear systems; interpolation and approximation; numerical differentiation and integration; numerical solution of ordinary differential equations. Texts: Chapra and Canale, *Numerical Methods for Engineers*, 8th ed., Parts 2, 3, 5, 6, 7; Burden, Faires, and Burden, *Numerical Analysis*. Reference: [MIT 18.330](#).

MATH 333 Computational Science and Engineering. Prerequisites: MATH 130, MATH 120. Applied linear algebra and the framework of equilibrium equations; difference methods for differential equations; Fourier methods; optimization and network models. Texts: Strang, *Computational Science and Engineering*; Burden, Faires, and Burden, *Numerical Analysis*. Reference: [MIT 18.085](#).

Probability and Statistics

MATH 340 Probability and Random Variables. Prerequisites: MATH 102. Combinatorial probability; discrete and continuous distributions; joint distributions; expectation and conditional expectation; moment generating functions; limit theorems. Texts: Ross, *A First Course in Probability*, 10th ed., ch. 1–8; DeGroot and Schervish, *Probability and Statistics*. Reference: [MIT 18.600](#).

MATH 341 Mathematical Statistics. Prerequisites: MATH 140. Sampling distributions; point estimation and maximum likelihood; confidence intervals; hypothesis testing and likelihood ratio tests; linear regression; nonparametric methods. Texts: Wackerly, Mendenhall, and Scheaffer, *Mathematical Statistics with Applications*, 7th ed., ch. 7–11; Hogg, McKean, and Craig, *Introduction to Mathematical Statistics*. Reference: [MIT 18.650](#).

MATH 342 Stochastic Processes. Prerequisites: MATH 310, MATH 340. Markov chains in discrete and continuous time; Poisson processes; martingales; Brownian motion; stationary processes. Texts: Grimmett and Stirzaker, *Probability and Random Processes*, 4th ed., ch. 6–8, 12, 13; Ross, *Stochastic Processes*. Reference: [MIT 18.445](#).

Discrete Mathematics

MATH 350 Combinatorics and Graph Theory. Prerequisites: MATH 110. Graphs, trees, and connectivity; matchings; coloring and planarity; enumeration, generating functions, and the principle of inclusion–exclusion; partitions. Texts: West, *Introduction to Graph Theory*, 2nd ed., ch. 1–5; Cameron, *Combinatorics: Topics, Techniques, Algorithms*. Reference: [MIT 18.314](#).

MATH 351 Algebraic Combinatorics. Prerequisites: MATH 350, MATH 120. Walks in graphs and the eigenvalues of adjacency matrices; the matrix-tree theorem; partially ordered sets; Young tableaux; enumeration under group action. Texts: Stanley, *Algebraic Combinatorics: Walks, Trees, Tableaux, and More*, 2nd ed.; Cameron. Reference: [MIT 18.212](#).

Computing

CS 320 Design and Analysis of Algorithms. Prerequisites: CS 120. Divide and conquer; greedy algorithms; network flow; amortized analysis; randomized algorithms; linear programming; NP-completeness and approximation algorithms. Texts: Cormen et al., *Introduction to Algorithms*; Kleinberg and Tardos, *Algorithm Design*. Reference: [MIT 18.410J](#) / [6.046J](#).

CS 360 Matrix Methods in Data Analysis and Machine Learning. Prerequisites: MATH 120. The singular value decomposition and low-rank approximation; principal component analysis; least squares and regularization; optimization by gradient descent; the linear algebra of neural networks. Texts: Strang, *Linear Algebra and Learning from Data*; Strang, *Introduction to Linear Algebra*. Reference: [MIT 18.065](#).

Course Catalog: Topics Courses and Seminars

The ten MATH 4xx courses correspond to MIT's undergraduate seminars. Each is an elective, assessed by problem sets; the student can take it as one of the two required seminars instead, and it still counts as an elective or free elective. Each carries 170 hours.

Taken as a seminar, the course follows MIT's format: the student chooses a topic within the area, studies it from primary sources, records a presentation, and writes an expository paper of about 10 pages, typeset in LaTeX, aimed at a reader who has completed the fixed courses. Texts are chosen per topic, with a bibliography in the paper.

Code	Title	Prerequisites	Topics at MIT	Reference
MATH 410	Topics in Analysis	MATH 210	Subject matter from current journals or books; topics vary from year to year	MIT 18.104
MATH 420	Topics in Algebra	MATH 221	Topics vary from year to year	MIT 18.704
MATH 425	Topics in Number Theory	MATH 221	Topics vary from year to year	MIT 18.784
MATH 440	Information Theory	MATH 140, MATH 120	Data compression, Shannon's theorems, and error-correcting codes	MIT 18.424
MATH 450	Topics in Discrete Mathematics	MATH 110, MATH 120	Combinatorics, graph theory, and discrete mathematics, from papers in recent literature	MIT 18.204
MATH 460	Topics in Topology	MATH 260	The fundamental group and covering spaces, and their relation to the theory of knots	MIT 18.904
MATH 465	Topics in Geometry	MATH 120, MATH 210	Subject matter from current journals or books; topics vary from year to year	MIT 18.994
MATH 470	Topics in Logic	MATH 210, MATH 120	Subject matter from current journals or books; topics vary from year to year	MIT 18.504
MATH 480	Topics in Theoretical Computer Science	CS 320	Topics vary from year to year	MIT 18.434
MATH 490	Project Laboratory in Mathematics	MATH 210, MATH 221	Guided research: projects in areas such as dynamical systems, number theory, algebra, and probability, each with a laboratory report, and an oral presentation	MIT 18.821

Prerequisites follow the [MIT Mathematics Department's list of communication-intensive subjects](#). Where MIT accepts any of several subjects, the program uses the one in its catalog; for MATH 490, MIT asks for any two mathematics subjects numbered 18.100 or higher, and the program uses its two fixed courses of that level. All ten subjects are listed in MIT's course catalog. Five references (18.424, 18.434, 18.704, 18.784, 18.994) point to an OpenCourseWare search because no course page was confirmed there; 18.504 points to the department's logic page.

Course Catalog: Free Electives

Four courses count only as free electives, because their reference subjects come from other MIT departments: electrical engineering and computer science, and physics. Any elective can also be counted as a free elective. Each carries 170 hours.

MATH 334 Optimization. Prerequisites: MATH 102, MATH 120. Convex sets and functions; linear programming and the simplex method; duality; unconstrained and constrained optimization; optimality conditions; algorithms for convex problems. Texts: Boyd and Vandenberghe, *Convex Optimization*, ch. 1–5, 9; Bazaraa, Jarvis, and Sherali, *Linear Programming and Network Flows*. Reference: [MIT 6.079](#).

CS 370 Introduction to Machine Learning. Prerequisites: MATH 102, MATH 120, CS 101. Formulating learning problems; linear models and gradient descent; regularization, overfitting, and generalization; classification and logistic regression; neural networks; clustering and dimensionality reduction; an introduction to reinforcement learning. Texts: Deisenroth, Faisal, and Ong, *Mathematics for Machine Learning*, ch. 8–12; James, Witten, Hastie, and Tibshirani, *An Introduction to Statistical Learning*, 2nd ed. Reference: [MIT 6.036](#).

PHYS 201 Vibrations and Waves. Prerequisites: MATH 102, PHYS 102. Free and damped oscillations; driven oscillators and resonance; coupled oscillators and normal modes; waves on strings, sound, and electromagnetic waves; Fourier analysis and wave packets; interference and diffraction. Texts: French, *Vibrations and Waves*; Georgi, *The Physics of Waves*. Reference: [MIT 8.03SC](#).

PHYS 202 Quantum Physics I. Prerequisites: PHYS 201, MATH 130. The experimental basis of quantum physics; wave functions and the Schrödinger equation; operators, expectation values, and the uncertainty principle; one-dimensional potentials, tunneling, and the harmonic oscillator; the hydrogen atom. Texts: Griffiths and Schroeter, *Introduction to Quantum Mechanics*, 3rd ed., ch. 1–4; French and Taylor, *An Introduction to Quantum Physics*. Reference: [MIT 8.04](#).

Instructional Model

Each course is studied from its primary text, with the reference course as the guide to pace and emphasis. There are no instructors; the student plans, studies, and checks their own work.

- **Primary text.** The main source of the course. The listed chapters define its scope; the supplementary text offers a second explanation or more advanced treatment.
- **Reference course.** Lecture notes, videos, and problem sets from MIT OpenCourseWare or another university. Where official solutions exist, they are the best way to check one's own work.
- **Weekly workload.** 170 hours over a 4-month term is about 10 hours per week per course, including problem sets. The program's app shows how many hours the courses in progress require each week and records the hours actually studied.
- **Due dates.** Each course's problem sets fall at weeks 4, 8, 12, and 15 of its term and the cumulative review at week 16, counted from the day the course starts.
- **Pace.** Terms move with the student: finishing early brings the next term forward, and falling behind pushes it back.

Assessment Policy

A course is complete when all its assignments are submitted fully solved; there are no grades. The standard is mastery: the student works, checks, and redoes each problem until the solution is correct and complete.

Course type	Assignments	Content
Regular course	Problem Sets 1–4	25 problems each, covering the material up to its due date
	Cumulative Review	10 problems across the whole course
Seminar	Presentation	A recorded talk on the chosen topic
	Paper	An expository paper of about 10 pages, typeset in LaTeX, with a bibliography

- **Sources of problems.** Problems come from the primary text, the reference course, or are written by the student. Problems generated with AI tools are allowed only after the student checks them against the text.
- **Submission.** Each assignment is a PDF of the statements and full solutions. The student stores it wherever they choose, makes the link public so that anyone with it can open the file, and pastes the link in the app, which keeps only the link.
- **Checking.** Where the reference course publishes official solutions, the student compares their work with them. Elsewhere, the student checks results by independent methods, special cases, or the text's answers.
- **Prerequisites.** Courses taken before their prerequisites are allowed, but the student is responsible for the gap.

Academic Integrity

Because the credential rests on published work, that work must be the student's own and honestly described.

- **Own solutions.** Every solution is written by the student. Consulting texts, notes, and discussions is encouraged; copying a solution is not.
- **AI tools.** AI may be used to explain concepts, suggest problems, or check reasoning. It may not write the solutions submitted as the student's work. Problems produced with AI are checked against the text before use.
- **Attribution.** Problems taken from a book or course are labeled with their source; adapted problems say so. Seminar papers cite every source they use.
- **Copyright.** Published files contain original or adapted problem statements with references, never reproduced pages of a textbook.
- **Accuracy.** Dates and completion records are not altered after the fact. A course that was skipped or left incomplete is shown as such.

Credential and Verification

The graduate issues their own diploma and publishes a verification page; the proof of the degree is the work itself, open to anyone.

- **Diploma.** Created, edited, and published by the graduate, outside the app; its wording and design are the graduate's choice. The verification page links to it. Stating that the degree is self-directed and not conferred by an accredited institution is recommended, to avoid any confusion with an official title.
 - **Verification page.** A web page file that the app generates at any time from what the student recorded, and that the student publishes on a web host of their choice (GitHub Pages, for example); the app does not host or publish it. It shows the student's name and one link, the start and finish dates, links to the diploma and this handbook, and every planned course by term, with its code, hours, dates, and the files of its assignments (or presentation and paper, for seminars).
 - **What a visitor can check.** That each course has its full set of work; that the work matches the course's scope in this handbook; and that the dates are consistent with the program's pace.
 - **What it does not certify.** The page proves that the work exists and is public. It does not prove who did the work or that it was done without help; readers judge that, as they would any portfolio.
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